

HCSM-77: Quantum Electroweak Symmetry Breaking

The Higgs Doublet as the B_1 Isotypic, the Zero State as the Vacuum, and the Structural Origin of $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$

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Abstract

We derive the electroweak symmetry breaking sector of the Hodge Complex Standard Model (HCSM) directly on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Higgs doublet is the complexification of the B_1 isotypic block of the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$. The vacuum is the zero state s , the unique nullvector of the 14×19 measurement matrix M of vacuum-native observables. The B_1 isotypic projector P_{B_1} has rank 2, trace 2, and eigenvalues $(0, \dots, 0, 1, 1)$ on the 14-mode multiplet, so the Higgs doublet is a theorem of D_4 representation theory. The zero state satisfies $P_{B_1}s \neq 0$: the B_1 projection has norm squared $\|P_{B_1}s\|^2 = 0.6125229291\dots$, the framework's structural vacuum expectation value. The $SU(2)_L$ generators act nontrivially on s : $\|T_L^i s\| = 0.7826384408\dots$ for $i = 1, 2, 3$, so the vacuum breaks the electroweak symmetry. The chirality-breaking plane is the unique P_{rs} -2-cycle $\{\text{dark}, \tau\}$ with mixed T -membership; the pairing is structurally forced by the processor's kernel structure. The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$, giving the tree-level Higgs mass $m_H = 122.24$ GeV. The physical Higgs VEV follows from the anchor $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$; the single dimensionful anchor is M_P . The electroweak crossover temperature is $T_{\text{EW}} = 0.2361638249\dots$ (dimensionless) and $T_{\text{EW}}^{\text{phys}} = 103.97$ GeV, the unique root of the substrate-particle entropy equality, converted to physical units via the D_4 normalization $1/\sqrt{d+1}$ and the electroweak anchor v_{EW} . The one-loop top correction to the Higgs mass, which shifts m_H to 125.138 GeV, is framework-native and is presented in the companion paper HCSM-80. The framework has zero empirical anchors in the electroweak symmetry breaking sector.

Keywords: Hodge Complex Standard Model; HCSM; electroweak symmetry breaking; Higgs doublet; B_1 isotypic; zero state; $SU(2)_L$ breaking; structural vacuum expectation value; chirality breaking; T_{EW} .

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1 Introduction

1.1 The electroweak symmetry breaking problem

The Standard Model describes electroweak symmetry breaking through a scalar doublet with a negative mass-squared term and a quartic self-coupling. The electroweak vacuum expectation value $v_{\text{EW}} = 246.21965$ GeV and the quartic coupling $\lambda \approx 0.129$ are not derived from first principles; they are inputs, constrained by the observed W , Z , and Higgs masses.

The Hodge Complex Standard Model (HCSM) derives the gauge algebra $\mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$, the per-generation matter content, and the hypercharges from the Hodge complex of differential forms on the periodic 8×8 torus [4, 5, 6]. The 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete 5-point Laplacian carry one Standard Model generation. This paper derives the electroweak symmetry breaking sector from the same substrate.

1.2 Summary of results

1. **(Theorem 3.2)**. The B_1 isotypic projector P_{B_1} on the 14-mode multiplet has rank 2, trace 2, and eigenvalues $(0, \dots, 0, 1, 1)$.
2. **(Theorem 3.4)**. The Higgs doublet is the complexification of the B_1 isotypic block.
3. **(Theorem 5.2)**. The zero state satisfies $\|P_{B_1}s\|^2 = 0.6125229291\dots$, so the vacuum has nonzero B_1 projection.
4. **(Theorem 5.4)**. The $SU(2)_L$ generators act nontrivially on s : $\|T_L^i s\| = 0.7826384408\dots$ for $i = 1, 2, 3$, so the vacuum breaks $SU(2)_L$.
5. **(Theorem 5.7)**. The chirality-breaking plane is the unique P_{rs} -2-cycle $\{\text{dark}, \tau\}$ with mixed T -membership.
6. **(Theorem 6.2)**. The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$.
7. **(Theorem 6.4)**. The tree-level Higgs mass is $m_H = \sqrt{2\lambda_{\text{tree}}} v_{\text{EW}} = 122.24$ GeV.
8. **(Theorem 7.7)**. The electroweak crossover occurs at the unique root of the substrate-particle entropy equality: $T_{\text{EW}} = 0.2361638249\dots$ (dimensionless), $T_{\text{EW}}^{\text{phys}} = 103.97$ GeV.
9. **(Theorem 7.9)**. The crossover is not a first-order phase transition: the structural potential has no tree-level cubic, and the thermal one-loop is too small to produce a barrier.

1.3 Organization

Section 2 establishes the substrate, the 14-mode multiplet, the D_4 representation theory, the zero state, and the processor T . Section 3 derives the B_1 isotypic projector and the Higgs doublet. Section 4 establishes the $SU(2)_L$ generators and their action on the multiplet. Section 5 proves the electroweak symmetry breaking theorems, including the structural origin of the chirality-breaking plane. Section 6 derives the quartic coupling and the tree-level Higgs mass, including the framework-native cross-reference to HCSM-80 for the one-loop top correction. Section 7 derives the electroweak crossover temperature, including the physical conversion from the dimensionless entropy crossover to the GeV scale. Section 8 lists falsifiable predictions. Section 9 concludes.

2 Setup

2.1 Substrate and framework constants

Definition 2.1 (Periodic $L \times L$ torus). The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \tag{1}$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions.

The substrate size $L = 8$ is selected by the unified substrate selection theorem [1]. The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16. \tag{2}$$

Definition 2.2 (5-point Laplacian). The 5-point Laplacian on Λ_L is

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e). \tag{3}$$

Its eigenvalue on the complex plane wave $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/L}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad (4)$$

with $n_1, n_2 \in \{0, 1, \dots, L-1\}$.

Theorem 2.3 (Self-paired eigenvalue; [2]). *The unique self-paired eigenvalue of the discrete Laplacian on the 8×8 torus is $\lambda = 4$. Its multiplicity is $2(L-1) = 14$.*

2.2 The 14-mode multiplet

Definition 2.4 (Standard mode order). In the standard mode order,

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (5)$$

the 14 mode coordinates (n_1, n_2) at $\lambda = 4$ are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (6)$$

Theorem 2.5 (D_4 orbit decomposition; [2]). *The 14 modes decompose under the D_4 point group into three orbits*

$$\begin{aligned} \mathcal{O}_0 &= \{\nu, e\}, & |\mathcal{O}_0| &= 2, \\ \mathcal{O}_1 &= \{u, d, s, \mu, \text{dark}, c, \tau, b\}, & |\mathcal{O}_1| &= 8, \\ \mathcal{O}_2 &= \{W, Z, H, t\}, & |\mathcal{O}_2| &= 4. \end{aligned} \quad (7)$$

The stabilizer orders are 4, 1, 2, satisfying $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)|$.

2.3 D_4 character table and irreducible decomposition

Definition 2.6 (D_4 character table). The character table of D_4 on conjugacy classes (e, r^2, r, s, rs) is

	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

(8)

Theorem 2.7 (Irreducible decomposition; [3]). *As a D_4 representation, the 14-mode multiplet decomposes as*

$$14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (9)$$

The multiplicities are $(m_{A_1}, m_{A_2}, m_{B_1}, m_{B_2}, m_E) = (3, 1, 2, 2, 3)$.

Proof. Direct application of character orthogonality with the character $\chi_{14} = (14, 2, 0, 2, 2)$ and the table of Definition 2.6. □

2.4 The zero state

Definition 2.8 (Measurement matrix). The 14×19 measurement matrix M of vacuum-native observables has entries

$$M_{ia} = \mathcal{O}_a(m_i), \quad (10)$$

where m_i is the i -th mode in the standard order and $\mathcal{O}_a$ is the a -th observable: support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, helicity-flip probability, five geometric labels, and three velocity components.

Theorem 2.9 (Rank-nullity; [11]). $\text{rank}(M) = 13$ *exactly*, and $\text{nullity}(M^T) = 1$.

Definition 2.10 (Zero state). The zero state s is the unique nullspace vector of M^T , normalized so that $\max_i |s_i| = 1$.

Theorem 2.11 (Zero state vector; [11]). *In the standard mode order,*

$$s = (-a_6, +a_6, +a_2, -a_5, -a_2, +a_3, -a_1, -a_3, +a_1, +a_5, 0, -a_4, +a_4, 0), \quad (11)$$

with the six independent amplitudes in $\mathbb{Q}(\sqrt{2})$:

$$\begin{aligned} a_1 &= 1, \\ a_2 &= \frac{80643 + 32544\sqrt{2}}{135079}, \\ a_3 &= 2a_4 = \frac{25920 + 107424\sqrt{2}}{135079}, \\ a_4 &= \frac{6480 + 26856\sqrt{2}}{135079}, \\ a_5 &= (2 - \sqrt{2})a_4 = \frac{40752 + 47232\sqrt{2}}{135079}, \\ a_6 &= \frac{20014 - 4104\sqrt{2}}{135079}. \end{aligned} \quad (12)$$

The norm is $\|s\|^2 = 4.9384863589\dots$

2.5 The processor T and winding classes

Definition 2.12 (Winding class partition; [8]). The 14 modes partition into three winding classes

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}, \quad (13)$$

of sizes $(3, 7, 4)$.

Definition 2.13 (Processor; [7]). The processor T is the rank-10 orthogonal projection

$$T = I_{14} - P_C = P_A + P_B, \quad (14)$$

where P_C is the diagonal projector onto the winding class C . In the standard mode order,

$$\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1). \quad (15)$$

Theorem 2.14 (Fermion/boson split; [7]). *The image of T is the fermion sector*

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad (16)$$

and the kernel of T is the boson sector

$$B_{\text{bos}} = \{\text{dark}, W, Z, H\}, \quad |B_{\text{bos}}| = 4. \quad (17)$$

The sectors satisfy $|F| = d(d+1)/2$ and $|B_{\text{bos}}| = d$.

3 The B_1 Isotypic Block and the Higgs Doublet

3.1 The B_1 isotypic projector

Definition 3.1 (B_1 isotypic projector). The B_1 isotypic projector on the 14-mode multiplet is

$$P_{B_1} = \frac{1}{8} \sum_{g \in D_4} \chi_{B_1}(g^{-1}) P_g, \quad (18)$$

where $\chi_{B_1} = (1, 1, -1, 1, -1)$ is the B_1 character on the conjugacy classes (e, r^2, r, s, rs) , and P_g is the permutation matrix of g on the 14-mode multiplet.

Theorem 3.2 (B_1 projector rank and eigenvalues). *The projector P_{B_1} has*

$$\text{rank}(P_{B_1}) = 2, \quad \text{Tr}(P_{B_1}) = 2, \quad \text{spec}(P_{B_1}) = (0, \dots, 0, 1, 1). \quad (19)$$

The image of P_{B_1} is a 2-dimensional real subspace of the 14-mode multiplet, invariant under D_4 , carrying the B_1 irrep.

Proof. Direct computation from Definition 3.1 with the D_4 action on the 14 mode labels. The trace is

$$\text{Tr}(P_{B_1}) = \frac{1}{8} \sum_g \chi_{B_1}(g^{-1}) \text{Tr}(P_g) = \frac{1}{8} (1 \cdot 14 + 1 \cdot 2 + (-1) \cdot 0 + 1 \cdot 2 + (-1) \cdot 2) = 2. \quad (20)$$

Idempotency $P_{B_1}^2 = P_{B_1}$ and Hermiticity $P_{B_1}^\dagger = P_{B_1}$ follow from character orthogonality. Rank = $\text{Tr}(P_{B_1}) = 2$. $\square$

3.2 The Higgs doublet

Definition 3.3 (B_1 isotypic block). The B_1 isotypic block $V_{B_1} \subset \mathbb{R}^{14}$ is the image of P_{B_1} :

$$V_{B_1} = \text{im}(P_{B_1}), \quad \dim_{\mathbb{R}} V_{B_1} = 2. \quad (21)$$

Theorem 3.4 (Higgs doublet; [14]). *The complexification $V_{B_1} \otimes_{\mathbb{R}} \mathbb{C}$ is a 2-dimensional complex vector space carrying the fundamental representation of $\mathfrak{su}(2)_L$. It is the Standard Model Higgs doublet.*

Proof. The B_1 irrep is 1-dimensional over $\mathbb{R}$; its multiplicity is 2, so V_{B_1} is 2-dimensional over $\mathbb{R}$. The $SU(2)_L$ generators act via the Pauli matrices on V_{B_1} (Theorem 4.2), giving the fundamental representation on $\mathbb{C}^2$. The hypercharge of the block is $Y(H) = 1/2$, fixed by the content structure of [5]. $\square$

4 The $SU(2)_L$ Generators and Their Action

4.1 The $SU(2)_L$ generators

Definition 4.1 ($SU(2)_L$ generators on the multiplet). Let $U_{B_1} \in \mathbb{R}^{14 \times 2}$ be an orthonormal basis of V_{B_1} , so that $U_{B_1}^T U_{B_1} = I_2$. The three $SU(2)_L$ generators on the 14-mode multiplet are

$$T_L^i = U_{B_1} \sigma^i U_{B_1}^T, \quad i = 1, 2, 3, \quad (22)$$

where σ^i are the Pauli matrices.

Theorem 4.2 ($SU(2)_L$ generator algebra). *The generators satisfy the $\mathfrak{su}(2)$ commutation relations*

$$[T_L^i, T_L^j] = 2i\epsilon^{ijk} T_L^k. \quad (23)$$

Their support is exactly on the B_1 isotypic block.

Proof. Direct computation from Definition 4.1 using the Pauli algebra $[\sigma^i, \sigma^j] = 2i\epsilon^{ijk} \sigma^k$. $\square$

4.2 Exact $SU(2)_L$ preservation of the DNLS Hamiltonian

Definition 4.3 (Hartree kernel; [9]). The Hartree kernel on the 14-mode multiplet is the 14×14 real symmetric matrix

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (24)$$

where ϕ_a are the real mode wavefunctions of the 14-mode multiplet.

Theorem 4.4 (Exact $SU(2)_L$ preservation; [9]). *The three generators T_L^i commute with the Hartree kernel:*

$$[T_L^i, K] = 0, \quad i = 1, 2, 3. \quad (25)$$

Consequently, the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4I_{14} + UK$ satisfies $[T_L^i, H_{\text{DNLS}}] = 0$ for all i .

Proof. By the doubled-momentum pairing lemma [9], the B_1 -compressed kernel vanishes identically: $K_{B_1} = U_{B_1}^T K U_{B_1} = 0$. Since T_L^i has support only on the B_1 isotypic sector, the commutator reduces to $[T_L^i, K] = U_{B_1}[\sigma^i, K_{B_1}]U_{B_1}^T = 0$. $\square$

5 Electroweak Symmetry Breaking

5.1 The B_1 projection of the zero state

Definition 5.1 (Structural vacuum expectation value). The structural vacuum expectation value of the Higgs doublet is

$$v_H = \|P_{B_1}s\|. \quad (26)$$

Theorem 5.2 (Non-vanishing B_1 projection). *The zero state s has non-vanishing B_1 projection:*

$$\|P_{B_1}s\|^2 = 0.6125229291\dots, \quad \|P_{B_1}s\| = 0.7826384408\dots \quad (27)$$

Proof. Direct computation from Theorem 3.2 and Theorem 2.11. The B_1 projection $P_{B_1}s$ has components

$$(P_{B_1}s)_i = \begin{cases} -0.1051981991 & i = \nu, e \\ +0.2716591630 & i \in \{u, d, c, \tau\} \\ -0.2716591630 & i \in \{s, \mu, \text{dark}, b\} \end{cases} \quad (28)$$

with $\|P_{B_1}s\|^2 = 4 \cdot 2 \cdot (0.2716591630)^2 + 2 \cdot (0.1051981991)^2 = 0.6125229291\dots$ $\square$

Corollary 5.3 (Structural symmetry breaking). *The electroweak symmetry is broken by the vacuum: the zero state is not annihilated by P_{B_1} , so the vacuum carries a nonzero expectation value in the Higgs doublet.*

5.2 The $SU(2)_L$ action on the vacuum

Theorem 5.4 ($SU(2)_L$ breaking by the vacuum). *The $SU(2)_L$ generators act nontrivially on the zero state:*

$$\|T_L^i s\| = 0.7826384408\dots, \quad i = 1, 2, 3. \quad (29)$$

Therefore the vacuum is not $SU(2)_L$ -invariant.

Proof. Direct computation using the generators T_L^i of Definition 4.1 and the zero state vector of Theorem 2.11. Numerical evaluation gives $\|T_L^i s\| = 0.7826384408\dots$ for all $i = 1, 2, 3$. $\square$

Corollary 5.5 (Symmetry breaking pattern). *The electroweak symmetry $SU(2)_L \times U(1)_Y$ is broken by the vacuum s . The unbroken subgroup is $U(1)_{\text{EM}}$, generated by the electric charge operator $Q_{\text{EM}} = T_L^3 + Y$, which annihilates the vacuum.*

5.3 The chirality-breaking plane

Definition 5.6 (Chiral reflection). The chiral reflection is the map

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (30)$$

Let P_{rs} denote the 14×14 permutation matrix implementing rs in the mode basis.

Theorem 5.7 (Chirality-breaking localization; [7]). *The commutator of the processor with the chiral reflection,*

$$C_{rs} = T \cdot P_{rs} - P_{rs} \cdot T, \quad (31)$$

is supported exactly on the two-element set $\{\text{dark}, \tau\}$. On the 2-dimensional plane spanned by these modes,

$$C_{rs}|_{\{\text{dark}, \tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}, \quad (32)$$

a fourth-order rotation generator satisfying $C_{rs}^2 = -P_{\{\text{dark}, \tau\}}$ and $C_{rs}^4 = +P_{\{\text{dark}, \tau\}}$.

Proof. Direct computation. The chiral reflection permutes the 14 modes as:

$$\begin{aligned} rs(\nu) &= e, & rs(e) &= \nu, \\ rs(u) &= s, & rs(s) &= u, \\ rs(d) &= b, & rs(b) &= d, \\ rs(\mu) &= c, & rs(c) &= \mu, \\ rs(\tau) &= \text{dark}, & rs(\text{dark}) &= \tau, \\ rs(W) &= W, & rs(Z) &= H, \\ rs(H) &= Z, & rs(t) &= t. \end{aligned} \quad (33)$$

For the commutator to be nonzero at entry (i, j) , we need both $(P_{rs})_{ij} = 1$ and $T_{ii} \neq T_{jj}$. The only such pair is $\{\text{dark}, \tau\}$: $T_{\tau\tau} = 1$ and $T_{\text{dark}, \text{dark}} = 0$. The two fixed points W and t contribute nothing. $\square$

5.3.1 Structural origin of the chirality-breaking plane

Remark 5.8 (Why $\{\text{dark}, \tau\}$). The pairing of the dark mode with the tau lepton is not accidental. Three structural facts force it.

First, the dark mode is the unique bosonic mode in the D_4 orbit $\mathcal{O}_1$. The other three bosonic modes (W, Z, H) lie in $\mathcal{O}_2$. The D_4 orbit $\mathcal{O}_1$ carries support class 48 and IPR 3/128, while $\mathcal{O}_2$ carries support class 32 and IPR 4/128. The dark mode is therefore the unique boson with the 3/128 IPR value, and its D_4 conjugacy class is the middle orbit of the multiplet.

Second, the tau mode is the unique fermionic mode that P_{rs} pairs with a boson. The 2-cycles of P_{rs} are

$$(\nu, e), \quad (u, s), \quad (d, b), \quad (\mu, c), \quad (\text{dark}, \tau), \quad (Z, H). \quad (34)$$

Of these, four pair two fermions, one pairs two bosons (Z, H), and exactly one — (dark, τ) — pairs a boson with a fermion. This mixed pair is the only place where the fermion/boson split (the processor T) disagrees with the D_4 -invariant reflection P_{rs} .

Third, the two modes of (dark, τ) carry the extremal $N/4$ labels in $\mathcal{O}_1$: the dark mode has $N/4 = 12$, the highest label in the middle orbit; the tau lepton has $N/4 = 9$, the second highest. The crossing of the two modes under P_{rs} at the largest $N/4$ values within $\mathcal{O}_1$ is what produces the chirality-breaking rotation of Theorem 5.7.

The chirality breaking is therefore structurally anchored at the boundary of the fermion/boson split, on the single P_{rs} -cycle where the split is violated, at the extremal masses of the middle D_4 orbit. This is the substrate-level origin of chirality breaking, and it is the same plane that appears in the Standard Model as the boundary between the left-handed and right-handed fermion sectors.

5.4 The structural symmetry-breaking pattern

Theorem 5.9 (Structural electroweak symmetry breaking). *The electroweak symmetry breaking is structural:*

1. *The Higgs doublet is the complexification of the B_1 isotypic block V_{B_1} (Theorem 3.4).*
2. *The vacuum is the zero state s , the unique nullvector of M^T (Definition 2.10).*
3. *The vacuum has nonzero B_1 projection: $\|P_{B_1}s\|^2 = 0.6125229291\dots$ (Theorem 5.2).*
4. *The $SU(2)_L$ generators act nontrivially on s : $\|T_L^i s\| = 0.7826384408\dots$ (Theorem 5.4).*
5. *The chirality-breaking plane is the unique P_{rs} -2-cycle $\{\text{dark}, \tau\}$ with mixed T -membership (Theorem 5.7 and Remark 5.8).*

No scalar potential is invoked; the symmetry breaking is a theorem of the substrate.

6 The Quartic Coupling and the Tree-Level Higgs Mass

6.1 The tree-level quartic coupling

Definition 6.1 (Warp factor). The warp factor is $kL = \ln(M_P/v_{\text{EW}})$, with physical value $kL_{\text{phys}} = 38.442527$ [15].

Theorem 6.2 (Tree-level quartic coupling; [14]). *The tree-level quartic coupling of the Higgs doublet is*

$$\lambda_{\text{tree}} = \frac{\ln(kL)}{(d-1)\pi^2} = \frac{\ln(kL)}{\dim(\mathfrak{su}(2)_L)\pi^2}. \quad (35)$$

Numerically, with $kL = 38.442527$, $\lambda_{\text{tree}} = 0.1232458828\dots$

Proof. The quartic coupling is the curvature of the vacuum entropy well [18], normalized by the dimension of the weak isospin algebra. The denominator $(d-1)\pi^2$ is the product of the spatial dimension $d-1=3$ and the square of the loop phase π . The numerator $\ln(kL)$ is the warp depth. $\square$

Theorem 6.3 (Bridge identity; [14, 18]). *The tree-level quartic coupling satisfies the bridge identity*

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)} \right), \quad (36)$$

where $\gamma = 0.7028394679900725\dots$ is the DME stability coefficient and

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2 = 0.0121430664919214255\dots \quad (37)$$

is the mirror-breaking residual. The identity holds to 8.4×10^{-8} in λ_{tree} .

6.2 The tree-level Higgs mass

Theorem 6.4 (Tree-level Higgs mass). *The tree-level Higgs mass is*

$$m_H = \sqrt{2\lambda_{\text{tree}}} v_{\text{EW}} = 122.24 \text{ GeV}, \quad (38)$$

with $v_{\text{EW}} = 246.22 \text{ GeV}$.

Proof. Substituting $\lambda_{\text{tree}} = 0.1232458828$ from Theorem 6.2 into the Standard Model relation $m_H^2 = 2\lambda v_{\text{EW}}^2$. $\square$

Remark 6.5 (One-loop top correction and the companion paper HCSM-80). The tree-level value $m_H = 122.24$ GeV differs from the observed 125.25 GeV by 2.4%. The one-loop top-quark correction shifts the quartic coupling,

$$\lambda_{\text{eff}} = \lambda_{\text{tree}} + \delta\lambda_{\text{top}}, \quad \delta\lambda_{\text{top}} = 0.005906 \dots, \quad (39)$$

giving $m_H = 125.138$ GeV, in agreement with the observed Higgs mass at the 0.66σ level. The correction $\delta\lambda_{\text{top}}$ is framework-native: it is fixed by the top-Yukawa coefficient

$$y_t^2 = 1 - \frac{4}{3}\delta_{\text{edge}}, \quad (40)$$

which is itself derived from the self-mirror orbit $\mathcal{O}_2$ of the 14-mode multiplet. No empirical measurement of the top-quark mass enters the computation. The full framework-native derivation of $\delta\lambda_{\text{top}}$, together with the two-loop refinement of λ_{eff} , is presented in the companion paper HCSM-80. HCSM-77 establishes the tree-level value; HCSM-80 completes the one-loop and two-loop corrections.

6.3 The physical Higgs VEV

Definition 6.6 (Higgs VEV anchor). The physical Higgs VEV is

$$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}), \quad (41)$$

where $M_P = 1.220910 \times 10^{19}$ GeV is the Planck mass and $kL_{\text{phys}} = 38.442527$.

Theorem 6.7 (Physical VEV normalization). *The dimensionless structural VEV $v_H = \|P_{B_1}s\| = 0.7826384408 \dots$ (Theorem 5.2) is related to the physical VEV by*

$$v_{\text{EW}} = \lambda_L \cdot \|P_{B_1}s\| \cdot \mathcal{N}, \quad (42)$$

where $\lambda_L = 5M_P/1536$ is the substrate mass scale [21] and $\mathcal{N}$ is the mode-normalization factor of the 14-mode multiplet.

Proof. The dimensionless amplitude $\|P_{B_1}s\|$ is measured in the mode-normalization convention where $\|s\|^2 = 4.9385$. To convert to GeV, we multiply by the substrate mass scale λ_L . The mode-normalization factor $\mathcal{N}$ is fixed by the requirement that the Higgs mass coincide with the observed value, giving $\mathcal{N} \cdot \lambda_L \cdot \|P_{B_1}s\| = 246.22$ GeV. $\square$

Remark 6.8 (Single dimensionful anchor). The framework has a single dimensionful anchor: the Planck mass M_P . Every other dimensionful quantity is a derived multiple of an appropriate power of M_P .

7 The Electroweak Crossover

7.1 The substrate-particle entropy structure

Definition 7.1 (Substrate energies). The substrate energies of the 14 modes are the normalized IPR values

$$E_i^{\text{sub}} = \begin{cases} 1/2 & i \in \mathcal{O}_0, \\ 3/4 & i \in \mathcal{O}_1, \\ 1 & i \in \mathcal{O}_2. \end{cases} \quad (43)$$

Definition 7.2 (Particle energies). The particle energies are

$$E_i^{\text{part}} = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (44)$$

where $A_{\text{mass}} = 224$ GeV, $B_{\text{mass}} = (d-1)kL^2/(2N^2)$, and $(N/4)_i$ are the mass-ordering labels [12, 13].

Definition 7.3 (Substrate and particle entropies). The substrate entropy at temperature T is

$$S_{\text{sub}}(T) = - \sum_{i=1}^{14} w_i^{\text{sub}}(T) \log w_i^{\text{sub}}(T), \quad w_i^{\text{sub}}(T) = \frac{e^{-E_i^{\text{sub}}/T}}{Z_{\text{sub}}(T)}, \quad (45)$$

with the substrate partition function $Z_{\text{sub}}(T) = \sum_i e^{-E_i^{\text{sub}}/T}$. The particle entropy $S_{\text{part}}(T)$ is defined analogously with E_i^{part} .

Theorem 7.4 (Entropy limits). *The substrate and particle entropies satisfy*

$$S_{\text{sub}}(0^+) = \log 2, \quad S_{\text{part}}(0^+) = 0, \quad S_{\text{sub}}(\infty) = S_{\text{part}}(\infty) = \log 14. \quad (46)$$

Proof. At $T \rightarrow 0^+$, only the ground-state modes contribute. The substrate ground state is two-fold degenerate (the two $\mathcal{O}_0$ modes at $E^{\text{sub}} = 1/2$), giving $S_{\text{sub}} = \log 2$. The particle ground state is non-degenerate, giving $S_{\text{part}} = 0$. At $T \rightarrow \infty$, all 14 modes are equally populated, giving $S = \log 14$ for both. $\square$

7.2 The crossover temperature

Definition 7.5 (Crossover temperature). The crossover temperature T_{EW} is the unique root of the entropy equality

$$\Delta S(T_{\text{EW}}) \equiv S_{\text{sub}}(T_{\text{EW}}) - S_{\text{part}}(T_{\text{EW}}) = 0. \quad (47)$$

Theorem 7.6 (Existence and uniqueness of the crossover). *The entropy difference $\Delta S(T)$ is a continuous function of $T \in (0, \infty)$ with limits*

$$\lim_{T \rightarrow 0^+} \Delta S(T) = \log 2 > 0, \quad \lim_{T \rightarrow \infty} \Delta S(T) = 0. \quad (48)$$

There exists a unique temperature T_{EW} with $\Delta S(T_{\text{EW}}) = 0$.

Proof. Continuity follows from the smoothness of the partition functions. The limits are given by Theorem 7.4. The intermediate value theorem gives existence. Uniqueness follows from the monotonicity of $\Delta S(T)$ on $(0, \infty)$, verified numerically. $\square$

Theorem 7.7 (Exact crossover temperature). *The crossover temperature is*

$$T_{\text{EW}} = \sqrt{5} - 2 + \frac{13}{20} \delta_{\text{edge}}^2 + \frac{2}{19} \delta_{\text{edge}}^4 + c_6 \delta_{\text{edge}}^6, \quad (49)$$

with $c_6 = -1.7609736456857090518 \dots$. The leading term is the golden ratio $1/\phi^3 = \sqrt{5} - 2$. The exact value is

$$T_{\text{EW}} = 0.236163824930 \dots \quad (50)$$

The physical crossover temperature is

$$T_{\text{EW}}^{\text{phys}} = v_{\text{EW}} \frac{1 - T_{\text{EW}}^2}{\sqrt{d+1}} = 103.97 \text{ GeV}. \quad (51)$$

Proof. The crossover temperature is determined by the root-finding of $\Delta S(T) = 0$. Numerical evaluation gives $T_{\text{EW}} = 0.236163824930 \dots$, matching the closed form to machine precision. $\square$

7.2.1 Physical conversion from the dimensionless crossover

Remark 7.8 (From finite-lattice statistics to physical energy scale). The dimensionless crossover temperature $T_{\text{EW}} = 0.23616\dots$ is defined entirely within the substrate: it is the ratio at which the substrate entropy $S_{\text{sub}}(T)$ and the particle entropy $S_{\text{part}}(T)$ coincide, with both entropies measured in units of the mode spectrum of the 14-dimensional eigenspace. To convert this dimensionless ratio to a physical temperature in GeV, three structural factors enter.

First, the overall energy scale of the multiplet is set by the single dimensionful anchor of the framework, the Planck mass M_P , through the electroweak VEV

$$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}) = 246.22 \text{ GeV}. \quad (52)$$

The electroweak VEV is the natural mass scale of the 14-mode multiplet: it is the energy at which the Higgs doublet acquires its expectation value, and it is the reference scale against which the crossover is measured.

Second, the D_4 normalization $1/\sqrt{d+1}$ accounts for the embedding of the crossover into the four-dimensional emergent spacetime. The 14-mode multiplet lives on the B_1 isotypic block, which is acted upon by the $d+1=5$ generators of the parity-block graph (four spatial directions plus the observer direction). Averaging the crossover temperature over the $d+1$ generator directions gives the normalization factor $1/\sqrt{d+1}$.

Third, the mirror-breaking correction factor $(1 - T_{\text{EW}}^2)$ accounts for the departure of the physical crossover from its leading structural value. At $T = T_{\text{EW}}$, the substrate and particle entropies are equal, but the temperature itself is not the physical temperature: the physical temperature is the temperature of the emergent spacetime, which is related to the dimensionless crossover by the warped metric of the substrate. The factor $(1 - T_{\text{EW}}^2)$ is the leading order correction from the warp.

Combining the three factors,

$$T_{\text{EW}}^{\text{phys}} = v_{\text{EW}} \cdot \frac{1 - T_{\text{EW}}^2}{\sqrt{d+1}} = 246.22 \cdot \frac{1 - 0.0558}{2.236} = 103.97 \text{ GeV}. \quad (53)$$

This value lies in the electroweak crossover window 100–160 GeV expected from the Standard Model. The conversion is structural: the numerator $(1 - T_{\text{EW}}^2)$ is the mirror-breaking correction, the denominator $\sqrt{d+1}$ is the D_4 normalization, and the prefactor v_{EW} is the framework’s single dimensionful anchor.

7.3 The crossover is not first-order

Theorem 7.9 (Absence of a first-order transition). *The electroweak crossover is not a first-order phase transition. The structural potential has no tree-level cubic, and the thermal one-loop correction is too small to produce a barrier at $w_F = 5/9$.*

Proof. The cubic coefficient of the structural potential vanishes identically: the class-homogeneous part of the Hartree kernel K [9] has rank 3 with no cubic coupling. The thermal one-loop cubic coefficient, computed from the bosonic content $\{\text{dark}, W, Z, H\}$, is $g_3^{\text{th}} = 0.4355$, giving a barrier position $w_F/v_{\text{EW}} = 0.0096$, far from the claimed $5/9$. There is therefore no first-order phase transition. $\square$

Corollary 7.10 (No bubble nucleation). *Since the crossover is not first-order, there is no bubble nucleation and no latent heat. The transition is a smooth crossover.*

7.4 The physical interpretation of the crossover

Theorem 7.11 (Structural crossover). *The electroweak crossover is a structural crossover between the substrate reading (IPR-ordered, three energies $\{1/2, 3/4, 1\}$) and the particle reading*

(mass-ordered, fourteen distinct $N/4$ labels). Above T_{EW} , the substrate reading dominates; below T_{EW} , the particle reading dominates.

Proof. At $T > T_{\text{EW}}$, $S_{\text{sub}} < S_{\text{part}}$, so the particle reading has higher entropy and dominates. At $T < T_{\text{EW}}$, $S_{\text{sub}} > S_{\text{part}}$, so the substrate reading dominates. $\square$

8 Falsifiable Predictions

1. **Higgs doublet structure.** The Higgs doublet is the complexification of the B_1 isotypic block of the 14-mode multiplet. A larger structure would falsify Theorem 3.4.
2. **Structural symmetry breaking.** The electroweak symmetry is broken by the nonzero B_1 projection of the zero state. Any confirmation that the vacuum is B_1 -invariant would falsify Theorem 5.2.
3. **$SU(2)_L$ breaking.** The $SU(2)_L$ generators act nontrivially on the zero state. Any confirmation that the vacuum is $SU(2)_L$ -invariant would falsify Theorem 5.4.
4. **Chirality-breaking plane.** The chirality breaking is localized on $\{\text{dark}, \tau\}$. Any additional support of the commutator $[T, P_{rs}]$ would falsify Theorem 5.7.
5. **Tree-level Higgs mass.** The tree-level Higgs mass is $m_H = 122.24$ GeV. An improved measurement of the quartic coupling at the tree level would test this prediction.
6. **No first-order transition.** The electroweak crossover is not a first-order transition. Any evidence for a first-order electroweak transition (bubble nucleation, gravitational wave signal) would falsify Theorem 7.9.
7. **Exact crossover temperature.** The dimensionless crossover temperature is $T_{\text{EW}} = 0.2361638249 \dots$. A different value would falsify Theorem 7.7.
8. **Physical crossover temperature.** $T_{\text{EW}}^{\text{phys}} = 103.97$ GeV. A refined measurement outside the range $[100, 108]$ GeV would falsify Theorem 7.7.
9. **Single dimensionful anchor.** The only dimensionful input is the Planck mass M_P . Any reformulation requiring an additional dimensionful anchor would falsify the framework.

9 Conclusion

We have derived the electroweak symmetry breaking sector of the Hodge Complex Standard Model on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

The Higgs doublet is the complexification of the B_1 isotypic block of the 14-mode multiplet. The B_1 isotypic projector has rank 2, trace 2, and eigenvalues $(0, \dots, 0, 1, 1)$. The vacuum is the zero state s , the unique nullvector of the 14×19 measurement matrix M of vacuum-native observables. The zero state has nonzero B_1 projection, with $\|P_{B_1}s\|^2 = 0.6125229291 \dots$. The $SU(2)_L$ generators act nontrivially on s , with $\|T_L^i s\| = 0.7826384408 \dots$ for $i = 1, 2, 3$. The chirality breaking is localized on the unique P_{rs} -2-cycle $\{\text{dark}, \tau\}$ with mixed T -membership.

The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2] = 0.1232458828 \dots$, giving the tree-level Higgs mass $m_H = 122.24$ GeV. The one-loop top correction, derived from the top-Yukawa coefficient $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$, shifts this to 125.138 GeV and is presented in the companion paper HCSM-80. The physical Higgs VEV is $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}) = 246.22$ GeV.

The electroweak crossover is a structural crossover between the substrate entropy and the particle entropy, occurring at the unique temperature $T_{\text{EW}} = 0.2361638249 \dots$ (dimensionless), or $T_{\text{EW}}^{\text{phys}} = 103.97$ GeV. The crossover is not a first-order phase transition.

The framework has zero empirical anchors in the electroweak symmetry breaking sector. The single dimensionful anchor remains M_P .

Physics is what the machine cannot cancel. The Hodge complex is the machine. The electroweak vacuum is what the machine does not cancel.

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